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ECE 414 Senior Design Project Proposal

Team number

Project Title: Multilingual Sign-to-Speech Communication Device: Edge-Deployed ASL
Recognition with Confidence-Gated Speech Output

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1. Abstract

We propose a portable, self-contained device that recognizes American Sign Language and speaks the result aloud in a selected language. A camera captures the signer, an on-board machine-learning model recognizes signs from a defined vocabulary, and the output is spoken only after the user confirms it on screen. The system abstains rather than voice an uncertain reading. A microphone and speech recognition display the hearing person's reply as text, making the exchange two-way. Components: camera, embedded graphics processing unit (GPU) computer, microphone, speaker, display. Intended for Deaf users, hospitals, schools and public services, where on-demand interpreters are costly or unavailable.

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3. Introduction

American Sign Language (ASL) is a complete natural language with its own grammar and vocabulary, distinct from English, and it is the primary language of about 500,000 Deaf and hard-of-hearing people in the United States [1]. When a Deaf ASL user needs to communicate with a hearing person who does not sign, a certified interpreter is usually needed, but one is often unavailable. Automatic translation could help fill this gap.

3.1 Background

Automatic sign language processing has advanced quickly, but no system yet translates ASL reliably enough for general real-world use [1]. Research is divided into two tasks. *Isolated sign language recognition* (ISLR) identifies a single sign from a fixed vocabulary. On the ASL Citizen dataset, which contains 83,399 videos of 2,731 signs from 52 signers, the best published baseline reaches 63.1% top-1 accuracy on signers not seen during training [2]. *Sign language translation* (SLT) converts continuous signing into full sentences. SLT models have traditionally been trained on *glosses*, written labels for each sign in a sentence, which are expensive to annotate and scarce for ASL [3].

Recent research is therefore moving toward gloss-free and self-supervised models. SignDino, for example, learns sign representations from video without any gloss or text labels [4]. Gan et al. similarly learn sign features without text supervision and use them for gloss-free translation [5]. A parallel trend is lightweight models that work on body and hand landmarks rather than on full video frames, which makes them small and fast enough for embedded hardware. MediaPipe is a widely used framework for extracting such landmarks [6]. The winning solution of the Google – Isolated Sign Language Recognition competition on Kaggle, a one-dimensional convolutional neural network (1D-CNN) combined with a transformer, is one example of a landmark-based recognizer [7]. Pose-only models have also been applied to continuous sign language recognition [8].

Because gloss-free continuous translation still requires large models and datasets, our project uses landmark-based isolated recognition over a limited vocabulary. This approach runs offline on an embedded graphics processing unit (GPU), can be evaluated sign by sign, and produces confidence scores that the device can use to abstain rather than guess.

3.2 Problem Statement

Deaf ASL users often cannot get an interpreter when they need one. In a recent survey of 32 U.S. Deaf and hard-of-hearing ASL users, participants most often wanted an interpreter but could not get one in medical settings, followed by professional and educational settings [1]. These are exactly the settings where interpreting is legally mandated, which shows that the problem is not only one of law but of availability [1]. Interpreters are costly, often must be booked in advance, and can take hours to arrive, especially in rural areas or at short notice [1].

Existing automated approaches also have three practical weaknesses. First, they typically output a translation even when uncertain, and a wrong translation can be worse than none, especially in high-stakes settings. Second, cloud-based systems need a reliable internet connection and stream video of the user to remote servers, which raises privacy concerns. Third, most are one-way: they convert signing to text but give the hearing person no way to reply without a second device.

Our device addresses each of these weaknesses. All processing runs locally on an embedded GPU, so the device works without a network connection and no video leaves it. A built-in microphone converts the hearing person's speech to on-screen text, so a single device supports a two-way conversation. The device speaks a sentence only after the user confirms it on screen, and when its confidence is low it shows a list of candidate signs instead of guessing. The device is intended to provide access when an interpreter is unavailable, such as at a service counter, a front desk, or while waiting for an interpreter to arrive. It is not a replacement for certified interpreters.

3.3 Objectives

The objective of this project is to design and build a portable prototype that lets a Deaf ASL user hold a short two-way conversation with a hearing person who speaks English, Turkish or Spanish, without an internet connection. The prototype is scoped to structured, everyday exchanges such as those at a service counter or front desk. Its technical goals, summarized in Table 1, are to recognize a vocabulary of 100–250 ASL signs at no less than 75% top-1 and 90% top-5 accuracy on signers not seen during training; to speak a confirmed sentence in the selected language within 1.2 s of the user's confirmation; to show the hearing person's spoken reply as English text; and to run entirely on an embedded GPU. The project also has a reliability objective: the device should abstain rather than voice a translation it is not confident in, and we will report its

accuracy on the sentences it does speak, not only its overall accuracy. These figures are design targets; measured results will be reported as the project progresses.

We chose the service-counter setting based on Deaf users' own priorities. Fig. 1 summarizes the survey ratings by setting. “Going into a business” received the highest willingness to use automatic translation, and every participant showed at least some willingness there; self-service kiosks and informal educational activities followed, while mental healthcare, education settings and formal personal events ranked lowest (Fig. 1(b)) [1]. Business settings also show a moderate unmet need for interpreters (Fig. 1(a)), so a device there would fill a real gap where users are most open to the technology. Participants also held their lowest accuracy expectations for kiosk-type exchanges and their highest for medical settings [1]. A service counter is therefore a realistic first target for a system with a vocabulary of a few hundred signs.

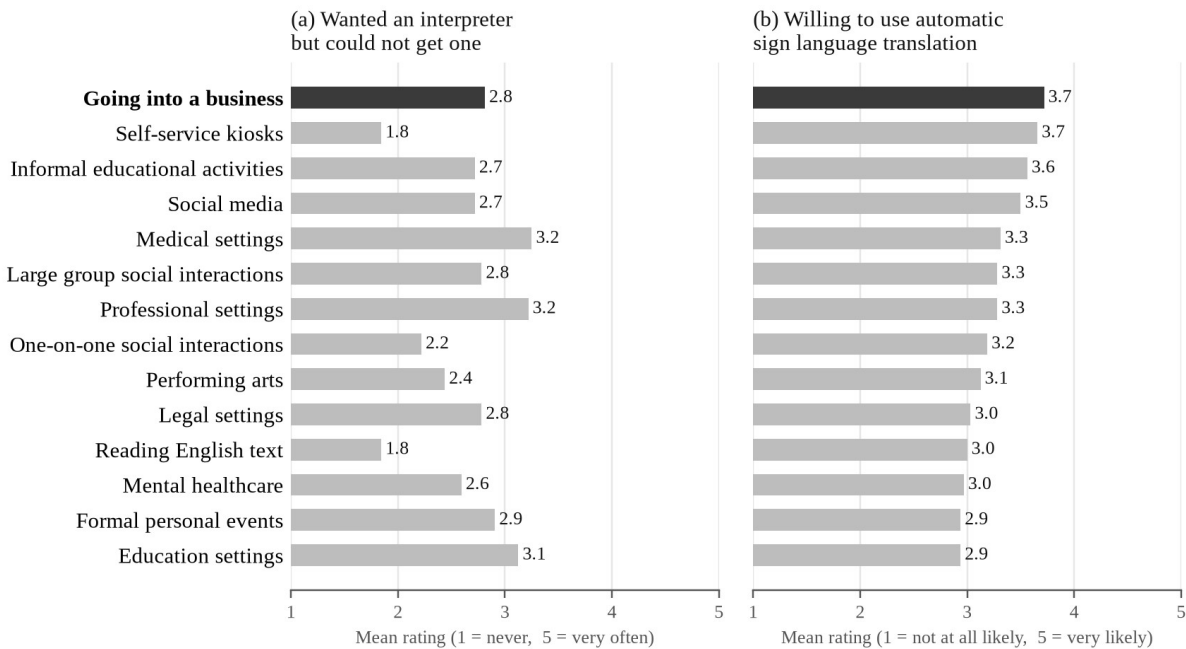


Fig. 1. Mean ratings by setting from a survey of 32 Deaf and hard-of-hearing ASL users: (a) how often they wanted a professional interpreter but could not get one; (b) how willing they would be to use automatic sign language translation. Settings are ordered by (b); the target setting of this project is highlighted. Data from Tran et al. [1]; means computed from the published response distributions.

Table 1. Design targets for the proposed device.

Parameter	Target	Rationale
Vocabulary	100–250 ASL signs, with fingerspelling as a fallback	Covers common service-counter exchanges
Top-1 accuracy (unseen signers)	$\geq 75\%$	Best published baseline is 63.1% on all 2,731 ASL Citizen signs [2]
Top-5 accuracy (unseen signers)	$\geq 90\%$	The correct sign should almost always appear in the candidate list
Latency	< 1.2 s from confirmation to speech	Keeps the conversation flowing
Output languages	English, Turkish, Spanish	Chosen by the user on the touchscreen
Connectivity	Fully offline	Works anywhere; no video leaves the device
Platform	NVIDIA Jetson Orin Nano	Embedded GPU in a portable enclosure

3.4 Proposed Approach

Fig. 2 shows how the device is used at a service counter. A single red-green-blue (RGB) camera and the touchscreen face the Deaf user, while the speaker and microphone face the hearing staff member. Body, hand and face landmarks are extracted from each camera frame with MediaPipe [6]. In natural signing, signs flow into each other with minimal pauses, and segmenting them automatically is still an open research problem [9]. The user therefore signs one sign at a time and returns the hands to a rest position between signs, and the device uses this rest pose to mark where each sign ends. A lightweight landmark-based model classifies each sign and outputs its five most likely labels with confidence scores; its design follows the winning Kaggle solution [7]. The model will be trained on the ASL Citizen dataset and evaluated on its signer-independent test split, as well as on recordings from our own test users [2]. Words outside the vocabulary, such as names, will be handled by a fingerspelling fallback that recognizes letters one at a time.

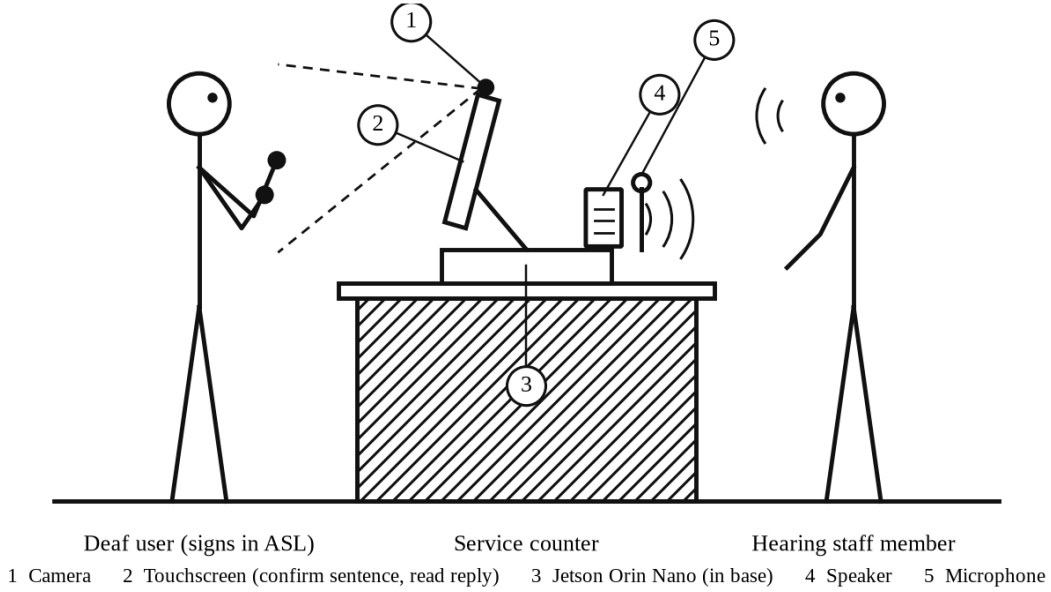


Fig. 2. The proposed device in use at a service counter (side view). The camera and touchscreen face the Deaf user; the speaker and microphone face the hearing staff member. All processing runs on the Jetson Orin Nano inside the base, without a network connection.

The recognized signs are ASL glosses, not English. ASL has its own word order and grammar, so a sequence such as STORE ME GO must become “I am going to the store.” A sentence builder performs this step, using templates for common phrases in the target setting and a small language model as a fallback, in line with recent work that applies pretrained language models to gloss-to-text translation [10]. The English sentence is shown on the touchscreen for the user to confirm or edit. Only confirmed sentences are translated into Turkish or Spanish with the No Language Left Behind (NLLB-200) translation model [11]. They are then spoken by Piper, an offline neural text-to-speech system [12]. In the reverse direction, a voice-activity detector starts listening when the hearing person speaks, and the microphone is muted while the device itself is speaking so that it does not transcribe its own voice. The reply is then transcribed and translated into English by the Whisper speech recognition model [13]. The text appears on the same touchscreen, so the hearing person needs no screen: they hear the device's speaker and speak into its microphone.

The device decides when to speak using the recognizer's confidence. If the confidence for a sign is above a threshold, the top sign is accepted; otherwise the touchscreen shows the five most likely candidates and the user picks one or signs again. Nothing is spoken until the user confirms the complete English sentence. This follows the idea of selective prediction, in which a model may

abstain on uncertain inputs to raise accuracy on the answers it does give [14]. We will evaluate this trade-off with risk–coverage curves and report accuracy separately for each test signer.

3.5 Current Trends

Three trends shape the field. First, research is moving from gloss-based pipelines toward gloss-free and self-supervised models that learn from large amounts of unlabeled video [4], [5]. Pretrained language models are also increasingly used for the language side of translation [10]. Second, models are moving onto consumer and embedded hardware. Landmark-based recognizers are small enough to run in real time on phones and single-board computers, which makes offline, privacy-preserving devices practical [6], [7]. Third, the field increasingly recognizes that sign language technology must be designed with Deaf communities rather than for them. Researchers have called for systems built around specific, real-world use cases with input from Deaf culture and linguistics experts [15]. Deaf and hard-of-hearing participants in recent studies also emphasize Deaf-led design and warn of harm when they are excluded from decisions [16]. Our project follows the second and third trends directly; it does not attempt gloss-free continuous translation, which still requires large models and datasets.

3.6 Existing Systems

No automatic ASL translation system is yet reliable enough for general use [1]. The best-known commercial effort, SignAll, combines multiple cameras with natural language processing to translate ASL into English text [17]. Patents show several other approaches, summarized in Table 2. Avodah's system uses multiple cameras with a depth sensor [18]. Sign-Speak's system recognizes signs from a single camera using cloud processing [19]. Bosch's system uses gloves with motion sensors [20], and an earlier Hitachi system uses data gloves or a camera with speech output [21]. The closest to our design is Sign-Speak's system, which also works from a single camera and, in one embodiment, lets the user choose among the three most likely translations [19]. As Fig. 3 shows, however, its recognition runs on a remote server, so it needs a network connection and sends the user's data off the device [19]. Our device differs from all four in combining single-camera input, fully offline processing on an embedded GPU, confidence-gated output and speech in several languages. It is also a form factor the Deaf community is open to: 69% of the survey participants were comfortable using a stand-alone system of this kind, compared with 38% for wearables [1].

Table 2. Comparison of patented sign language translation systems with the proposed device.

System	Input	Processing	Output
Avodah [18]	Multiple cameras and a depth sensor	On device and in the cloud	Text, speech or a signing avatar; two-way
Sign-Speak [19]	Single camera	Remote server (requires a network connection)	Text in a target language; optional menu of the three most likely translations
Bosch [20]	Gloves with motion sensors	External processor	Text or audio in one or more languages
Hitachi [21] (expired)	Data gloves or camera	Local computer	Text or synthesized speech
<i>Proposed device</i>	Single RGB camera and microphone	On device, fully offline	Speech in English, Turkish or Spanish after user confirmation; reply shown as English text

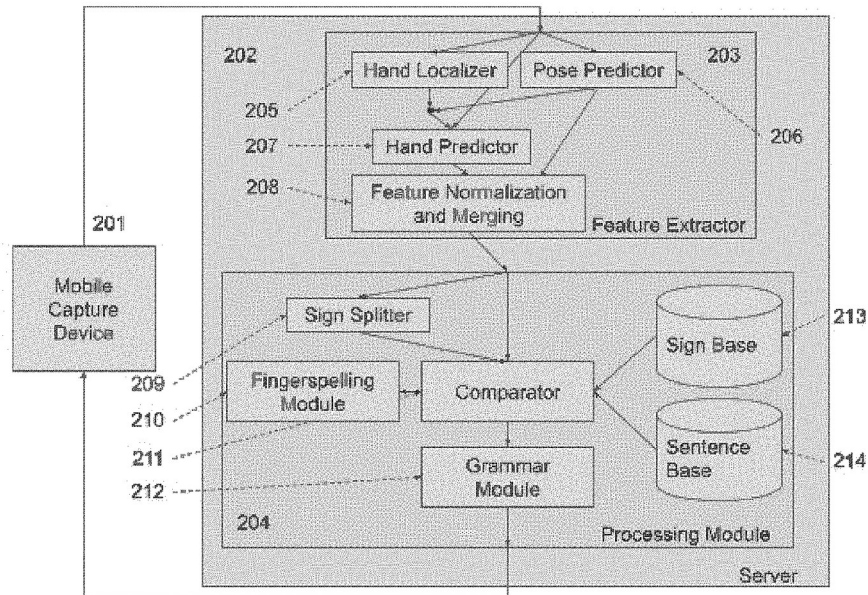


Fig. 3. Translation architecture in the Sign-Speak patent. A mobile capture device sends data to a remote server, where features are extracted and signs are matched against a sign and sentence base. Reproduced from U.S. Patent Appl. Pub. 2022/0327961 A1 [19].

3.7 Ethics, Economics and Sustainability

Surveys of Deaf communities also identify the ethical risks of this technology. More than 80% of the participants in the U.S. survey were concerned about hearing people profiting from ASL, systems missing ASL grammatical features, automatic translation weakening the legal right to other accommodations, and poorer performance for some users than others [1]. A 2025 global

survey found the same pattern: about 80% of Deaf respondents were concerned about hearing people making money from sign language technology, and about 70% about the lack of Deaf people leading or contributing to it (Fig. 4) [16]. We address these concerns directly. The device is presented as a way to communicate when an interpreter is unavailable, not as a substitute that could justify withholding one. Its limits are stated plainly: it recognizes individual signs and does not capture facial grammar or continuous signing. Accuracy will be reported for each test signer, not only as an average. We will seek input from Deaf ASL users when choosing the vocabulary and testing the device, as recommended by previous studies [15], [16]. Because video never leaves the device, the design also avoids the privacy risk of streaming a user's image to a remote server.

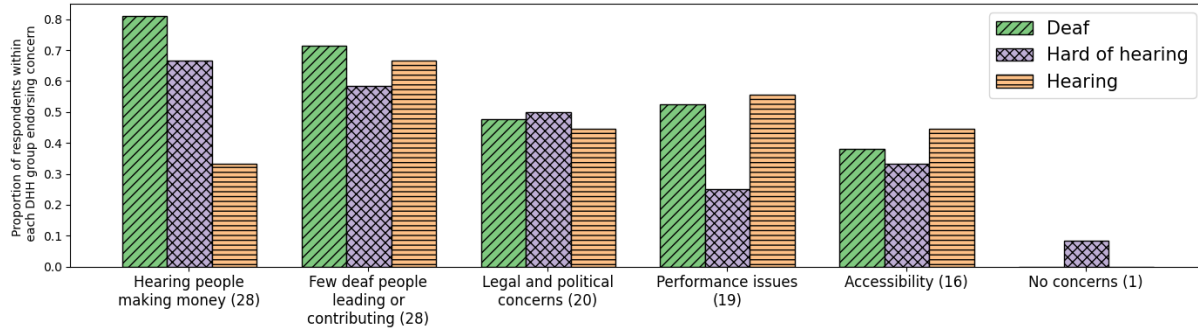


Fig. 4. Concerns about sign language technologies among Deaf, hard-of-hearing and hearing respondents in a global survey; numbers in parentheses are the number of mentions. Reproduced from Atwell et al. [16] under the Creative Commons Attribution 4.0 (CC BY 4.0) license.

Economically, certified interpreters are costly and often must be scheduled in advance [1]. Our device is built from off-the-shelf components (an embedded GPU module, a camera, a microphone, a speaker and a display) and uses open-source models, so once built it has no per-use service fees or cloud costs. A detailed cost estimate is given in Section 9. For sustainability, the device runs on a low-power embedded module rather than a desktop GPU or cloud servers, which keeps energy use low and avoids continuous data transmission. Standard, replaceable components make the device repairable and upgradable, and the software can be moved to newer hardware instead of the device being discarded.

4. Technical Goals

This section defines the three major technical goals of the project. Each goal corresponds to one part of the system shown in Fig. 2 and has measurable targets, so that the prototype can be validated objectively during ECE 416. The goals build on each other: Goal 1 produces recognized signs, Goal 2 turns them into a confirmed spoken sentence and carries the reply back, and Goal 3 makes the complete system run on portable hardware without a network connection. Table 3 summarizes the targets and how each will be verified, and Table 4 lists the team members responsible for each goal.

4.1 Goal 1: Sign Recognition with Abstention

The first goal is to recognize a vocabulary of 100–250 ASL signs from the camera image, and to know when the recognition is not reliable. The vocabulary will be chosen for the service-counter setting, covering greetings, common requests, questions about prices and directions, numbers and basic responses. To make sure the selected signs are the ones Deaf users actually need, the vocabulary will be reviewed with Deaf ASL users before training begins.

Recognition works on body and hand landmarks rather than raw video, which keeps the model small enough for the embedded hardware. MediaPipe extracts the landmarks from each frame [6], and the user returns the hands to a rest position between signs so that the device can tell where each sign ends. A lightweight landmark-based classifier, following the design of the winning Kaggle solution [7], will be trained on the ASL Citizen dataset [2] and outputs the five most likely signs with a confidence score for each.

The confidence scores are as important as the predictions themselves, because the device uses them to decide whether to accept a sign or to ask the user. A calibration step will adjust the scores so that, for example, signs reported with 90% confidence are correct about 90% of the time. The targets are at least 75% top-1 and 90% top-5 accuracy on signers not seen during training, measured on the ASL Citizen signer-independent test split and on recordings from our own test users. In addition, we will report the accuracy on the signs the device accepts together with the share of signs it accepts, using risk–coverage analysis [14]. Accuracy will be reported separately for each test signer, so that any signer for whom the system performs poorly is visible rather than hidden in an average.

4.2 Goal 2: Confirmed Multilingual Speech and Two-Way Conversation

The second goal is to turn the recognized signs into a correct spoken sentence in the language the user selects, and to carry the hearing person's reply back as text. Recognized signs are ASL glosses, so a sentence builder first produces natural English. It uses templates for frequent

service-counter phrases, such as “I would like ____” or “How much is ____?”, and falls back to a small language model for other sign sequences [10].

The English sentence is shown on the touchscreen, where the user can confirm it, replace a word with another candidate, delete a word, or sign again. Only a confirmed sentence is passed on. If Turkish or Spanish is selected, the sentence is translated with NLLB-200 [11]; if English is selected, translation is skipped. Piper then speaks the sentence with a voice for the selected language [12]. In the reverse direction, a voice-activity detector starts recording when the hearing person speaks, and the microphone is muted while the device is speaking. Whisper transcribes the reply and translates it into English [13], using the selected language as a hint to improve accuracy, and the text is shown on the touchscreen.

The main target for this goal is a latency of less than 1.2 s from the user's confirmation to the start of speech, which keeps the conversation flowing. Translation quality will be checked on a fixed test set of service-counter phrases, with each Turkish and Spanish output rated by native speakers. The accuracy of the reply channel will be measured as the word error rate of the transcription on recorded test replies.

4.3 Goal 3: Offline Embedded Integration

The third goal is to run the complete system on an NVIDIA Jetson Orin Nano, which the team will purchase, in a portable enclosure with a camera, touchscreen, microphone and speaker. All processing must happen on the device without any network connection, which is what allows the device to protect users' privacy and to work in places without internet access.

Running several neural networks at the same time on an embedded module is the main engineering challenge of the project. The landmark extractor, sign recognizer, sentence builder, translation model, text-to-speech system and speech recognizer all share the module's 8 GB of memory and its graphics processor. The models will therefore be converted to optimized runtimes such as ONNX Runtime or TensorRT, and the smallest model size that meets the accuracy targets will be chosen for each component. The targets are landmark extraction and recognition at no less than 15 frames per second during live use, all models fitting within the available memory, and a complete conversation working with networking disabled. We will also measure the device's power draw in watts, both idle and in use, to choose between battery and wall power.

4.4 Team Responsibilities

Each team member will lead one area and support another, so that every goal has at least two people who understand it and the project does not depend on a single person for any part.

Shared tasks, including the schedule, reports, presentations and user testing, are coordinated by the project-wide lead. The assignments will be finalized by the team and are listed in Table 4.

Table 3. Measurable targets and verification methods for the technical goals.

Goal	Target	How it will be verified
1. Sign recognition	$\geq 75\%$ top-1 and $\geq 90\%$ top-5 accuracy on unseen signers, for 100–250 signs	ASL Citizen signer-independent test split and recordings from our own test users
1. Sign recognition	Accuracy on accepted signs reported together with the share of signs accepted	Risk–coverage curves over a range of confidence thresholds
2. Language and speech	< 1.2 s from the user's confirmation to the start of speech	Timestamps logged on the device for every test sentence
2. Language and speech	Turkish and Spanish output judged acceptable for service-counter phrases	A fixed test set of phrases rated by native speakers
2. Language and speech	Hearing person's reply shown as readable English text	Word error rate of the transcription on recorded test replies
3. Embedded integration	≥ 15 frames per second for landmark extraction and recognition	Profiling on the Jetson Orin Nano during live use
3. Embedded integration	All models fit within 8 GB of memory; full conversation works offline	Memory monitoring and end-to-end tests with networking disabled
3. Embedded integration	Power draw measured to select the power supply	Power measurement in watts during idle and active use

Table 4. Team responsibilities for the technical goals.

Area	Lead	Support	Main tasks
Goal 1: Sign recognition	To be assigned	To be assigned	Vocabulary selection with Deaf ASL users, landmark extraction, rest-pose segmentation, model training, confidence calibration, accuracy evaluation
Goal 2: Language and speech	To be assigned	To be assigned	Sentence builder, touchscreen interface, translation, text-to-speech, speech recognition, latency measurement
Goal 3: Embedded integration	To be assigned	To be assigned	Hardware purchase and setup, enclosure, deployment and optimization on the Jetson Orin Nano, frame-rate and power measurement
Project-wide	To be assigned	To be assigned	Schedule and milestones, reports and presentations, user testing with Deaf ASL users, documentation

5. Projected Community Impact and Relevant Ethical Issues

5.1 Community Impact

The main beneficiaries of this project are Deaf ASL users, a group whose need for interpreters often goes unmet [1]. In everyday situations such as buying something at a store, asking a question at a bank or checking in at a front desk, a Deaf person today often has to write notes, rely on a family member to interpret, or simply leave without the information they needed. The device lets them communicate in their own language and on their own, with the hearing person hearing a spoken sentence instead of reading a note. Because the user confirms every sentence before it is spoken, they stay in control of what is said on their behalf.

The device also serves multilingual communities. Because the spoken output can be in English, Turkish or Spanish, a Deaf customer can communicate directly with staff or family members who are more comfortable in Turkish or Spanish than in English. This matters in diverse regions such as New Jersey, where many families and workers speak a language other than English at home.

Businesses and public services benefit as well. Under the Americans with Disabilities Act (ADA), businesses and nonprofits that serve the public must communicate effectively with people who have communication disabilities [22]. For long or complex conversations, a qualified interpreter remains the right solution, and the device does not replace one. For short, routine exchanges for which an interpreter is rarely available at short notice, the device gives staff a practical way to communicate. Because it runs offline, it can also be used in rural areas and in buildings without reliable internet access.

The design also has environmental benefits. Running on a low-power embedded module instead of cloud servers reduces energy use and avoids the continuous upload of video. Because the device is built from off-the-shelf, replaceable parts and open-source software, it can be repaired and upgraded rather than discarded, and its software can be moved to newer hardware at the end of the module's life.

5.2 Ethical Issues

We follow the IEEE Code of Ethics, which asks engineers to hold paramount the safety, health and welfare of the public, to protect the privacy of others, to avoid harm, to treat all persons fairly, and to be honest and realistic in stating claims [23]. The most important ethical issues for this project are discussed below, and Table 5 summarizes all of them together with our response to each.

The first issue is the risk of replacing interpreters. In the U.S. survey, 88% of Deaf participants were concerned that automatic translation could weaken the legal right to an interpreter [1]. A device like ours could be misused by an organization as a cheaper substitute in situations where a qualified interpreter is required. We will therefore describe and label the device as an aid for situations where no interpreter is available, and state this both in its documentation and on its start screen.

The second issue is harm from wrong translations. A misrecognized sign can change the meaning of a sentence, and in medical, legal or financial situations the consequences can be serious. Our design addresses this in two ways: the device abstains and shows candidate signs when it is unsure, and nothing is spoken until the user has read and confirmed the English sentence. High-stakes settings remain outside the intended use of the prototype.

The third issue is fairness. Sign recognition systems can work less well for some people, for example because of differences in signing style, handedness, skin tone or lighting. We will test with signers who differ in these respects, report accuracy for each signer rather than only an average, and publish any limits we find. Privacy and consent are closely related: the camera captures users' faces and bodies, so all processing stays on the device and no video or audio is stored or transmitted. We use consent-based training data such as ASL Citizen [2], and any recordings we make ourselves will be collected with written consent.

Finally, Deaf involvement is the issue that Deaf communities themselves rank highest. Our team is hearing, and both surveys cited in Section 3 show that the strongest concerns are hearing people profiting from ASL and sign language technology being built without Deaf input [1], [16]. We will therefore contact the ASL community at NJIT before choosing the vocabulary, invite Deaf signers to test the prototype, credit their contributions, and adjust the design based on their feedback. Any testing with people will follow NJIT's requirements for research involving human participants.

Table 5. Ethical issues and how the project addresses them.

Issue	Risk	Our response
Replacing interpreters	Organizations could use the device to avoid providing a required interpreter, weakening Deaf users' legal rights	Present and label the device as an aid for when no interpreter is available, in all documentation and on its start screen
Wrong translations	A mistranslation in a medical, legal or financial exchange can cause harm	Nothing is spoken without user confirmation; the device abstains when unsure; high-stakes settings are out of scope
Unequal performance	Accuracy may be lower for some signers because of signing style, handedness, skin tone or lighting	Test with diverse signers and report accuracy for each signer, along with known limits

Issue	Risk	Our response
Privacy	The camera captures users' faces and bodies, and the microphone records the reply	All processing stays on the device; no video or audio is stored or transmitted
Data consent	Training data come from real people	Use consent-based datasets such as ASL Citizen; obtain written consent for any recordings we make
Deaf involvement	Hearing-led projects that profit from ASL are the community's top concern	Involve Deaf ASL users in vocabulary selection and testing, and credit their contributions
Scope and honesty	Users may expect full ASL translation	State plainly that the device recognizes individual signs and does not capture facial grammar
Accessibility	Replies shown as English text may be hard for users less fluent in written English	Use short sentences and an adjustable font size; record this as a known limitation

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